Data Sheet

EXPERIMENT NO. 08

Title: - Simulation of Summing Amplifier using Op-Amp (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Inverting Summing Amplifier (general case): -

$Vo = - R_f \times [(V1/R1) + (V2/R2) + (V3/R3)]$

- V1, V2, V3 = Input voltages
- R1, R2, R3 = Input resistors
- R_f = Feedback resistor

Equal Input Resistors (R1 = R2 = R3 = R): -

$Vo = - (R_f / R) \times (V1 + V2 + V3)$

Unity-Gain Summing (R_f = R): -

$Vo = - (V1 + V2 + V3)$

Observation Table: -

Sl. No.	V1 (V)	V2 (V)	R1 (kΩ)	R2 (kΩ)	Rf (kΩ)	Vo (Calculated)	Vo (Measured)

Calculation: -

Vo (Calculated) = $- R_f \times [(V1/R1) + (V2/R2)]$ = _____

% Error between Calculated and Measured V_O = _____ %

Conclusion: -

Teacher's Sign and Date

Data Sheet

EXPERIMENT NO. 09

Title: - Simulation of Logarithmic and Anti-Logarithmic Amplifier using Op-Amp (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Logarithmic Amplifier (diode/transistor in feedback path): -

The diode current, $I_D = I_S\,(e^{(V_i/V_T)} - 1) \approx I_S \cdot e^{(V_i/V_T)}$ for $V_D \gg V_T$

Since the input current $I = V_i / R$ flows through the diode (virtual ground at op-amp input):

$V_O = -V_D = -V_T \cdot \ln(V_i / (R \cdot I_S))$

- V_T = thermal voltage (≈ 26 mV at room temperature)
- I_S = diode reverse saturation current
- R = input resistor, V_i = input voltage

Anti-Logarithmic (Exponential) Amplifier (diode/transistor at input): -

$I_D = I_S \cdot e^{(V_i/V_T)}$

$Vo = -R_f \cdot I_D = -R_f \cdot I_S \cdot e^{(V_i/V_T)}$

- R_f = feedback resistor, V_i = input voltage applied across the diode

Observation Table — Logarithmic Amplifier: -

Sl. No.	Input Voltage, Vi (V)	Output Voltage, Vo (Measured) (V)	Vo (Calculated) (V)

Observation Table — Anti-Logarithmic Amplifier: -

Sl. No.	Input Voltage, Vi (V)	Output Voltage, Vo (Measured) (V)	Vo (Calculated) (V)

Calculation: -

Log Amp: $V_O = -V_T \ln(V_i / (R \cdot I_S)) =$ _____

Anti-log Amp: $V_O = -R_f \cdot I_S \cdot e^{(V_i/V_T)} =$ _____

Conclusion: -

Teacher's Sign and Date

Data Sheet

EXPERIMENT NO. 10

Title: - Simulation of Phase Shift Oscillator using Op-Amp — Study of Frequency of Oscillation by Varying Component Values (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Barkhausen Criterion for Sustained Oscillation: -

Loop Gain, $|Av \cdot \beta| = 1$ and total phase shift around the loop = 360° (or 0°)

- The op-amp (inverting amplifier) contributes 180° phase shift
- The three-section RC feedback network contributes the remaining 180° (60° per section) at the frequency of oscillation

Frequency of Oscillation (three equal RC sections): -

$f = 1 / (2\pi RC \sqrt{6})$

- R = resistance and C = capacitance of each RC section (all equal)

Minimum Gain Condition for Oscillation: -

Feedback factor of a 3-section RC network, $\beta = 1/29$

Hence the amplifier must provide a gain, $|Av| = Rf / R1 \geq 29$

Observation Table: -

Sl. No.	R (kΩ)	C (nF)	Rf (kΩ)	R1 (kΩ)	f (Theoretical) Hz	f (Practical / Measured) Hz

Calculation: -

f (Theoretical) = 1 / (2π RC $\sqrt{6}$) = _____

% Error between Theoretical and Practical Frequency = _____ %

Conclusion: -

Teacher's Sign and Date